

## Accomplishments - What was done? What was learned?

### \* What are the major goals of the project?

[UW/GMU] With the advent of the digital era, the importance of natural language processing (NLP) technology for disaster response has increased dramatically, from triaging the need for immediate assistance based on text messages after the Haiti earthquake in 2010, to translating vast amounts of important data related to COVID-19 in order to fight the associated infodemic. The goal of the project is to propose a concrete path towards building language technologies for crisis/disaster response that will be useful for everyone. Our roadmap requires the development of crucial infrastructure to support technological solutions that enable and support communications in crisis scenarios. The envisioned infrastructure will consist of datasets applicable to carefully curated crisis/disaster scenarios tied to prototypical language scenarios that are necessary for communicating with affected populations. The infrastructure will facilitate research and advance knowledge in machine translation, spoken language translation, speech recognition, classification, summarization as well as enable their application in disaster/crisis scenarios.

[UW] This project is a collaborative endeavor between the University of Washington (UW) and George Mason University (GMU). As part of the broader resource development initiative, the UW team will focus specifically on the creation of Social Media (SM) datasets. These datasets will comprise messages curated from platforms such as Twitter, enriched with detailed annotations and summarizations, as outlined in the grant proposal and described in the following section.

[GMU] This project is a collaborative endeavor between the University of Washington (UW) and George Mason University (GMU). As part of the broader resource development initiative, the GMU team will focus specifically on the creation of speech translation (ST) datasets that facilitate cross-cultural communication in crisis scenarios. These datasets will comprise speech corpora, transcribed and translated, centered around the crisis domain, and curated in collaboration with language data providers and community members, as outlined in the project proposal.

### \* What was accomplished under these goals and objectives (you must provide information for at least one of the 4 categories below)?

#### Major Activities:

[UW] Our major activities in Y1 are as follows:

- We conducted a comprehensive survey to date (2020 to 2024) on the use of language technologies (LT) in crisis preparedness and response (LT4CPR), focusing on how social media data is leveraged during emergencies. We followed the PRISMA guidelines and systematically reviewed 355 relevant peer-reviewed papers out of 1,256 papers initially selected for review that were published between 2020 and 2024, examining corpus creation, system development, and real-world applications.

- We worked on designing the “situation report” for crisis preparedness and response. A situation report is a structured summary with additional information such as links to the source and explanation of the information in the report. We also worked on designing a pipeline for creating such reports that includes both automatic systems and human annotation.
- We conducted a systematic review of commercial Machine Translation (MT) systems to determine if their quality and language coverage is sufficient to be used in the next pandemic, or any crisis for that matter. Using the TICO-19 data set (Anastasopoulos et al 2020), we examined MT coverage and quality for the 36 languages covered in TICO-19, examining quality of the two major MT providers, namely Google and Microsoft, but also of two commercial Large Language Models (LLMs), namely GPT and Gemini.
- The interdisciplinary nature of our envisioned infrastructure requires input from several disciplines and subfields. To bring together like-minded LT researchers to discuss LT4CPR related projects and plan for future ones, we, in collaboration with the GMU team, organized a Birds-of-Feather (BoF) meeting at ACL 2025, held at Vienna, Austria in July 2025.
- A good venue for LT4CPR related research is the Workshop on NLP for Positive Impact. We presented our survey paper at the workshop and discussed with the workshop organizers about using LT4CPR datasets for next year’s shared tasks.

[GMU] Our major activities in Y1 are as follows:

- In this first year, we focused on establishing the major partners that would facilitate data collection – the major part of the resource creation. We identified ClearGlobal Inc (formerly Translators without Borders) as a potential partner, and established a first contractual agreement for the creation of corpora for 10 under-resourced languages. ClearGlobal is a non-profit focusing on providing language services in disaster relief and crisis response scenarios, who closely works with other non-profits (such as the Red Cross) or governmental organizations across the globe.
- We co-organized several community engagement activities (organizing Birds-of-a-Feather activities in two conferences), as well as engaging with indigenous language communities through invited lectures in appropriate venues.

### **Specific Objectives:**

[answer]

### **Significant results:**

[UW] Results:

- Our survey on the use of social media data in crisis preparedness and response was published and presented at the 4th Workshop on NLP for Positive Impact (<https://sites.google.com/view/nlp4positiveimpact>).
- We formed a concrete plan for creating situation reports based on social media data. We will start implementing the pipeline soon and hope to use the initial set of annotation data for a shared task next year.

- We organized the Birds-of-Feather meeting at ACL 2025 and received a lot of feedback from the attendees.

[GMU] - We developed an initial set of speech translation systems and speech translation metrics tailored to low-resource languages, with a focus on indigenous languages representing vulnerable populations. Our systems yield promising results, obtaining the top spots in related shared tasks.

#### **Key outcomes or Other achievements:**

[UW] Key outcomes:

- Key findings from our LT4CPR survey work included existing studies' strong reliance on English and Twitter data, a lack of multimodal and multilingual resources, and limited engagement with humanitarian stakeholders. The paper highlights challenges and outlines future research directions, including broader linguistic coverage, development of multimodal corpora, and closer collaboration with crisis response communities. This work directly supports the objectives of this grant by advancing the understanding of how LT can enhance crisis informatics and disaster response.
- Our survey paper on commercial MT systems found that coverage for majority languages—what we call pivots, since we can pivot from these languages to most languages of the world—is well covered by existing tools, but quality varies significantly across lower resourced languages in the set (but whose speaker populations still number in the millions to hundreds of millions).
- We conducted a thorough review of existing studies on summarization and situation report generation in the crisis informatics field. We found that most of the reports or summaries generated by the proposed models usually do not have structures. The lack of structure could make it difficult to gather useful information or trace the source of information in the generated outputs.

[GMU]

- contracting for the bulk of the data creation is complete, and we are expecting to start releasing our first data pack (as part of organizing a shared task) in the next months.
- We organized or participated in four community building events.

#### **\* What opportunities for training and professional development has the project provided?**

[UW] The project supported the training of one PhD student at GMU funded as a Graduate Research Assistant.

Both the LT4CPR survey paper and the MT survey work were conducted by the PIs, the RA, and a group of master students at the UW. The students were co-authors on the two survey papers.

[GMU] The project supported the training of one PhD student at GMU funded as a Graduate Research Assistant. In addition, the PI along with his PhD student focused on training of indigenous communities on AI and its potential. In particular, they participated by presenting in the following events:

- Towards Language Technologies for Indigenous Peoples, International Translation Day 2025. Sept 30th, 2025. Organized by Translation Commons, Virtual.
- AI for Indigenous Languages, Indigenous Digital Activism Summit. Global Voices. Mexico. March 14, 2025.
- Innovation in Indigenous NLP and ASR, Indigenous in AI Workshop at NeurIPS, 2024, Canada. Dec 12th. 2024
- "Research topics in NLP: LRL, Culture and Formal Languages". Jornadas de Jóvenes Lingüistas ("Young Linguist Day"), University of Buenos Aires. Sep, Wed 24th. 2025.
- MT Crisis Cookbook: Updates & Challenges. Pennsylvania, Penn State University. 12th Mid-Atlantic Student Colloquium on Speech, Language and Learning (MASC-SLL 2025). April, 2025.

**\* Have the results been disseminated to communities of interest? If so, please provide details.**

[UW] Yes.

We organized birds-of-a-feather sessions in one conference: at ACL 2025 in Vienna, Austria – the major conference for computational linguistics.

Our LT4CPR survey paper was published and presented at the 4th Workshop on NLP for Positive Impact in conjunction with ACL 2025 in Vienna. We also participated in the workshop's discussion on its plan for 2026 and offered to organize a shared task using our social media datasets.

[GMU] Yes, but more events are planned for years Y2 and Y3. We organized birds-of-a-feather sessions in one conference: at ACL 2025 in Vienna – the major conference for computational linguistics. We also connected with the translators and interpreters' community through participation in the International Translation Day activities. Last, we reached out to under-resourced language communities such as latin american indigenous ones through participation in events in Mexico City and Buenos Aires, promoting the use of AI for indigenous languages.

The reception of our project in all above settings has been overwhelmingly positive.

**\* What do you plan to do during the next reporting period to accomplish the goals?**

[UW] In Y2, we plan to implement the pipeline for situation report generation, create our first dataset, and release it to the community. will also propose to co-organize the NLP for Positive Impact workshop, and potentially organize a shared task using that dataset.

[GMU] In Y2, we will release our first data packs of disaster-related speech translation corpora. We are aiming to cover 4 languages. We will organize a shared-task as part of IWSLT to promote interest and encourage research on this task. We will also propose to co-organize the NLP for Positive Impact workshop, and potentially organize a shared task there as well.



## Products

**[UW] The LT4CPR survey paper**

## Participants & Other Collaborating Organizations - Who has been involved?

### \* What individuals have worked on the project?

[UW]

- Fei:
  - Dr. Xia and Lewis led the UW team in conducting the survey on LT4CPR and designing the pipeline for generating situation reports based on social media data.
  - Dr. Xia supervised Haotian Zhu, the RA on the project, and several master students working on the project as their milestone projects for the master degree.
  - The UW team collaborated with the GMU team in organizing a birds-of-a-feather meeting on LT4CPR at ACL 2025.
  - Dr. Xia attended Dr. Lewis' seminar on LT4CPR in spring 2025 and contributed to class discussion and provided feedback to students on their team projects on LT4CPR.
- Will:
  - Dr. Xia and Lewis led the UW team in conducting the survey on LT4CPR, presenting the work at the NLT4PositiveImpact Workshop and designing the pipeline for generating situation reports based on social media data.
  - Dr. Lewis taught a graduate seminar entitled *Language Technologies for Crisis Preparedness and Response (LT4CPR)*, and was focused on a survey of the research in various areas of LT4CPR. The projects from the seminar and the followup work led to a survey paper on whether the current MT technology is sufficient for crisis response.
  - Dr. Lewis presented the LT4CPR survey work at the NLT4PositiveImpact Workshop and the Birds-of-Feather meeting at ACL 2025.
- Haotian:
  - Participated in conducting the survey on LT4CPR and designing the pipeline for generating situation reports based on social media data.

### \* What other organizations have been involved as partners?

[UW] None other than GMU

**Were other collaborators or contacts involved? If so, please provide details.**

[UW] None.

Impact - What is the impact of the project? How has it contributed?

\* What is the impact on the development of the principal discipline(s) of the project? 

 NSF Specific

**Describe how findings, results, techniques that were developed or extended, or other products from the project made an impact or are likely to make an impact on the base of knowledge, theory, and research and/or pedagogical methods in the principal disciplinary field(s) of the project.**

[UW] same as GMU

[GMU] The project aims to catalyze research in language technologies for crisis response and preparedness. Even though we are at the beginning, we aim to effectively boost this into an active research sub-area for both NLP and crisis informatics.

\* What is the impact on other disciplines?

**Describe how the findings, results, or techniques that were developed or improved, or other products from the project made an impact or are likely to make an impact on other disciplines.**

[answer]

\* What is the impact on the development of human resources? 

 NSF Specific

**Describe how the project made an impact or is likely to make an impact on human resource development in science, engineering, and technology.**

[answer]

\*What was the impact on teaching and educational experiences? 

**Describe how the project made an impact or is likely to make an impact on teaching and educational experiences.**

[UW] In the Spring Quarter 2025, one of the PIs, Professor Lewis, taught a graduate seminar entitled *Language Technologies for Crisis Preparedness and Response (LT4CPR)*, and was focused on a survey of the research in various areas of LT4CPR. The seminar was attended by 10 graduate students who, for the seminar grade, worked on projects that took on issues in the LT4CPR space: multimodal data processing over social media feeds in crises, using large language models to effectively translate emergency alerts into languages not served by existing



alert infrastructure, and reviewing the readiness of existing machine translation tools for future pandemic response. For the latter project, the students did a comprehensive analysis of existing commercial machine translation tools in the context of pandemic response across a set of languages where a crisis might strike in the future. Shortly after the term ended, these students continued working on this project, drawing in two additional students to work on it. The end result is a paper that we are working on, which was detailed previously, namely, *Is MT Ready for the Next Crisis or Pandemic?*

[GMU] This project has informed part of the PI's teaching of Advanced NLP. In particular, some of the proposed semester-long projects for MSc or PhD correspond to objectives of this proposal, such as developing culture-aware translation models.

**\* What is the impact on physical resources that form infrastructure?**

**Describe ways, if any, in which the project made an impact, or is likely to make an impact, on physical resources that form infrastructure, including physical resources such as facilities, laboratories, or instruments.**

[UW] N/A

**\* What is the impact on institutional resources that form infrastructure?**



**Describe ways, if any, in which the project made an impact, or is likely to make an impact, on institutional resources that form infrastructure,**

[UW] same as GMU

[GMU] The project aims to establish a series of workshops and shared tasks around the sub-area of "LT4CPR". We are on track with proposing the first shared tasks and co-organizing such workshops in Y2.

**\* What is the impact on information resources that form infrastructure?**



**Describe ways, if any, in which the project made an impact, or is likely to make an impact, on information resources that form infrastructure,**

[UW] We are building several datasets to facilitate research on LT4CPR. We plan to make our first release in Y2.

[GMU] The project will release several corpora to facilitate research on LT4CPR. We have not done so in Y1, but we are on track to do so in Y2 of the project.

**\* What is the impact on technology transfer?**



**Describe ways in which the project made an impact, or is likely to make an impact, on commercial technology or public use.**

[UW] Our GMU/UW team is collaborating with organizations such as ClearGlobal, a non-profit that is directly involved in crisis relief, response, and preparedness, and which has a dedicated branch on actually developing the language technologies to support personnel in such situations. We hope that our collaboration will help them in refining their operations.

[GMU] We have established a close collaboration -which, however, just started- with ClearGlobal, a non-profit that is directly involved in crisis relief, response, and preparedness, and which has a dedicated branch on actually developing the language technologies to support personnel in such situations. Our project will have a direct impact on their operations, once completed.

**\* What is the impact on society beyond science and technology?** 

**Describe how results from the project made an impact, or are likely to make an impact, beyond the bounds of science, engineering, and the academic world.**

[UW] Same as GMU

[GMU] We hope that, once our project is completed, a large community of scientists will be working towards building LT4CPR, which in turn should benefit in actual crisis situations, once such AI tools get deployed for real-world scenarios.

**\* What percentage of the award's budget was spent in a foreign country?** 

 NSF Specific

**Describe what percentage of the award's budget was spent in foreign country(ies) for this reporting period. If more than one foreign country was involved, identify the distribution of funding between the foreign countries.**

[UW] None

## Changes/ Problems

### \* Changes in approach and reasons for change

[answer]

### \* Actual or Anticipated problems or delays and actions or plans to resolve them

[UW] Due to the new contract for UW graduate students, the RA's stipend and benefit rate have gone up much faster than what was expected at the time of our proposal submission. To deal with this financial gap, we plan to convert some quarters of RA support to hourly students.

It has become more difficult to access social media data due to the change of social media platforms' policies. Given this, we plan to use some existing, publicly available social media datasets such as CrisisNLP and CrisisLex as the base for the first release of our social media dataset. Meanwhile, we will explore more ways of accessing more recent social media data.

[GMU] It took about a 8 months for George Mason and ClearGlobal to agree on the verbiage of the contractual agreement for the creation and release of speech data in 10 under-resourced languages. This will likely end up pushing the project back by about a year, as we were not able to release our first data packs in Y1, but we are certain that we will start producing the infrastructure in Y2, as outlined in our proposal.

### \* Changes that have significant impact on expenditures

[answer]

### \* Significant changes in use or care of human subjects

 NSF Specific

[answer]

### \* Significant changes in use or care of vertebrate animals

 NSF Specific

[answer]

### \* Significant changes in use or care of biohazards

[answer]

\* Has there been a change in your primary performance site location from the originally proposed? If so, please provide the location of your new primary performance site and reason for the change in location.

[answer]

## Special Requirements

**Respond to any special reporting requirements specified in the award terms and conditions, as well as any award specific reporting requirements.**

[answer]